Qinyuan Wu

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Research Interests

I study what large language models actually know – how they memorize and organize knowledge – and whether that knowledge translates into reliable retrieval and action, particularly in tool-using agents. A single thread runs through my work, from **memorization** and **latent-knowledge estimation** to the **reliability of agentic tool use**. I draw on ideas from human memory to make these systems more cognitively grounded.

Education

CS@Max Planck & Saarland University — Oct 2022 – Present
Ph.D. in Computer Science
Advisors: Prof. Krishna P. Gummadi (MPI-SWS) & Prof. Muhammad Bilal Zafar (Ruhr-Universität Bochum)

University of Electronic Science and Technology of China (UESTC) — Sep 2018 – Jun 2022
B.Sc. in Mathematics-Physics Fundamental Science
Yingcai Honors College (top 2% selected)

Publications

*Author in **bold**; ★ marks papers where I am (equal-)first author.*

Conferences

C7. **Temporal Context Reinstatement Drives Episodic-Like Order Memory in Long-Context Language Models.**
[ICML 2026] *The Forty-Third International Conference on Machine Learning*
Mathis Pink, Vy A. Vo, **Qinyuan Wu**, Jianing Mu, Javier S. Turek, Uri Hasson, Kenneth A. Norman, Sebastian Michelmann, Alexander Huth, Mariya Toneva

C6. **Fine-tuning vs. In-context Learning in Large Language Models: A Formal Language Learning Perspective.**
[ACL 2026] [Main] *The 64th Annual Meeting of the Association for Computational Linguistics*
Bishwamittra Ghosh, Soumi Das, Till Speicher, **Qinyuan Wu**, Mohammad Aflah Khan, Deepak Garg, Krishna P. Gummadi, Evimaria Terzi

C5. **Characterizing Web Search in The Age of Generative AI.**
[ACL 2026] [Findings] *The 64th Annual Meeting of the Association for Computational Linguistics*
Elisabeth Kirsten, Jost Große Perdekamp, **Qinyuan Wu**, Mihir Upadhyay, Krishna P. Gummadi, Muhammad Bilal Zafar

C4. ★ **The Algorithmic Self-Portrait: Deconstructing Memory in ChatGPT.**
[WWW 2026] [Oral] *The ACM Web Conference*
Abhisek Dash*, Soumi Das*, Elisabeth Kirsten*, **Qinyuan Wu***, Sai Keerthana Karnam, Krishna P. Gummadi, Thorsten Holz, Muhammad Bilal Zafar and Savvas Zannettou
* Equal contribution

C3. **In Agents We Trust, but Who Do Agents Trust? Latent Source Preferences Steer LLM Generations.**
[ICLR 2026]*The Fourteenth International Conference on Learning Representations*
Mohammad Aflah Khan, Mahsa Amani, Soumi Das, Bishwamittra Ghosh, **Qinyuan Wu**, Krishna P. Gummadi, Manish Gupta, Abhilasha Ravichander.

C2. ★ **Rote Learning Considered Useful: Generalizing over Memorized Knowledge in LLMs.**
[ICLR 2026]*The Fourteenth International Conference on Learning Representations*
Qinyuan Wu, Soumi Das, Mahsa Amani, Bishwamittra Ghosh, Mohammad Aflah Khan, Krishna P. Gummadi, Muhammad Bilal Zafar.

C1. ★ **Towards Reliable Latent Knowledge Estimation in LLMs: Zero-Prompt Many-Shot Based Factual Knowledge Extraction.**
[WSDM 2025]*ACM International Conference on Web Search and Data Mining*.
Qinyuan Wu, Mohammad Aflah Khan, Soumi Das, Vedant Nanda, Bishwamittra Ghosh, Camila Kolling, Till Speicher, Laurent Bindschaedler, Krishna P. Gummadi, Evimaria Terzi.

Preprints

P4. ★ **To call or not to call: A framework to assess and optimize LLM tool calling.**
arXiv preprint, 2026.
Qinyuan Wu, Soumi Das, Mahsa Amani, Arijit Nag, Seungeon Lee, Krishna P. Gummadi, Abhilasha Ravichander, Muhammad Bilal Zafar

P3. **Position: Episodic Memory is the Missing Piece for Long-Term LLM Agents.**
arXiv preprint, 2025.
Mathis Pink, **Qinyuan Wu**, Vy Ai Vo, Javier Turek, Jianing Mu, Alexander Huth, Mariya Toneva.

P2. **Assessing Episodic Memory in LLMs with Sequence Order Recall Tasks.**
arXiv preprint, 2024.
Mathis Pink, Vy A. Vo, **Qinyuan Wu**, Jianing Mu, Javier S. Turek, Uri Hasson, Kenneth A. Norman, Sebastian Michelmann, Alexander Huth, Mariya Toneva.

P1. **Understanding Memorisation in LLMs: Dynamics, Influencing Factors, and Implications.**
arXiv preprint, 2024.
Till Speicher, Mohammad Aflah Khan, **Qinyuan Wu**, Vedant Nanda, Soumi Das, Bishwamittra Ghosh, Krishna P. Gummadi, Evimaria Terzi.

Workshops

W3. **Revisiting Privacy, Utility, and Efficiency Trade-offs when Fine-Tuning Large Language Models.**
The International Association for Safe & Ethical AI second annual conference (IASEAI 26)
Soumi Das, Camila Kolling, Mohammad Aflah Khan, Mahsa Amani, Bishwamittra Ghosh, **Qinyuan Wu**, Till Speicher, Krishna P. Gummadi.

W2. **Rethinking Memorization Measures in LLMs: Recollection vs. Counterfactual vs. Contextual Memorization.**
ICML 2025 Workshop: Impact of Memorization on Trustworthy Foundation Models
Bishwamittra Ghosh, Soumi Das, **Qinyuan Wu**, Mohammad Aflah Khan, Krishna P. Gummadi, Evimaria Terzi, Deepak Garg.

W1. **Testing Memory Capabilities in Large Language Models with the Sequential Ordered Recall Task.**
LatinX in AI @ NeurIPS 2024.
Mathis Pink, Vy A. Vo, **Qinyuan Wu**, Jianing Mu, Javier S. Turek, Uri Hasson, Kenneth A. Norman, Sebastian Michelmann, Alexander Huth, Mariya Toneva.

Before Ph.D.

J1. **Exponential negation of a probability distribution.** *Soft Computing*, 2022. **Qinyuan Wu**, Yong Deng, Neal Xiong.

Talks

IASEAI annual conference 2026
Rote Learning Considered Useful: Generalizing over Memorized Knowledge in LLMs
Data Science for Humanity Group, MPI-SP (Germany) 2025
Rote Learning Considered Useful: Generalizing over Memorized Knowledge in LLMs
AI and Society Group, Ruhr Universität Bochum (Germany) 2024
Reliable Knowledge Estimation of LLMs

Teaching Experience

Efficient Training of Large Language Models (TA) Winter 2025–26, Saarland University
Systems for Large (Language) Models (TA) Winter 2023–24, Saarland University

Academic Service

Reviewer: ARR 2025 (October cycle), ICML 2026, FAccT 2026, COLM 2026

Technical Skills

Model Training & Serving: Trained and deployed LLMs with open-source distributed frameworks; optimized inference using vLLM and SGLang; fine-tuned models (LoRA) and integrated HuggingFace pipelines on Slurm-managed multi-GPU clusters; performed data preprocessing (Pandas, NumPy, Polars).

Programming & Tools: Python, C, Bash, Git, Linux.
Languages: English (fluent), Chinese (native), German (beginner)

Awards

First-class Scholarship (Top 10%) 2020–2021
Innovation & Entrepreneurship Project: Excellent (Top 10%, State Level, 20K RMB) 2020–2021
Innovation & Entrepreneurship Project: Excellent (Top 10%, Univ. Level) 2019–2020
Second-class Scholarship (Top 20%) 2019–2020

References

Krishna P. Gummadi MPI-SWS, Scientific Director
gummadi@mpi-sws.org
Muhammad Bilal Zafar Ruhr University Bochum, Professor
bilal.zafar@rub.de